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Republic of the Philippines

Cagayan State University

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COLLEGE OF ENGINEERING

LABORATORY EXPERIMENT 1.3:

Logistic Population Models with Harvesting

In partial fulfilment for the requirements of the course

Differential Calculus

(Math 211E)

By: GROUP 5

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Abstract

This experiment used two models of harvesting which are the constant and periodic harvesting. In constant harvesting system, the values of a_1 are set from 0 to 0.25 to limit the system. Beyond the 0.25 value of a_1 the population of fish decreases until it approaches extinction. For periodic harvesting, since it is non-autonomous equation, the effect of the initial condition when $a=a_1$ is a decrease in the population of fish per season of harvesting. When $a= a_2$ there is a rapid drop of fish population.

I. Introduction

Fishery management is the balance between harvesting and its ecological implications. Worldwide, fish populations are in various states; some are healthy and some are depleted due to environmental and human factors. The population conditions greatly impact the dynamics of the system. In fishery management, it is important to fish in such a way that a species is sustainable and not in danger of becoming endangered or going extinct.

Two logistic growth models have been used namely constant harvesting and periodic harvesting. Logistic growth model is appropriate for population growth of animal when overcrowding and competition resources are taken into consideration.

Mathematical model have been used widely to estimate the population dynamics of animals for so many years as well as the human population dynamics. According to Aanes et al. (2002), the most important for successful management of harvested populations is that harvesting strategies are sustainable, not leading to instabilities or extinctions and produces great results for the year with little variation between the years. Therefore, it can supply the market demand throughout the year.

Harvesting has been an area under discussion in population as well as in community dynamics (Murray 1993). Malthus was the first to formulate theoretical treatment of population dynamics in 1798 and Verhulst formed the Malthus theory into a mathematical model called the logistic equation that led to nonlinear differential equation (Alan 1992). Dubey et al. (2003), John et al. (2005) and Biswajit et al. (2007) agreed that it is a need to develop an ecologically suitable strategy for harvesting any renewable resource. Jing and Ke (2004a, 2004b), and Li and Wang (2010) studied the optimal harvesting policies as their management intention over a random harvesting time perspective.

According to Idels and Wang (2008), constant harvesting is where a fixed numbers of fish were removed each year, while periodic harvesting is usually thought of a sequence of periodic closure and openings of different fishing grounds. Harvesting has been considered a factor of stabilization,

destabilization, and improvement of mean population levels, induced fluctuations, and control of non-native predators (Michel 2007). Further reference on harvesting strategies can be found in Cooke and Nusse (1987) and Ludwig (2006).

Objectives

Generally, this experiment aims to understand the logistic growth model with harvesting. Specifically it aims to:

1. Determine the effect on the fish population for constant harvesting $a=a_1$

2. Determine the effect on the fish population for constant harvesting $a=a_2$

3. Describe the parameter used in the model and how it is used for constant and periodic harvesting

II. Methodology

A. Logistic Growth with Constant Harvesting

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a$$

Where: P = size of the fish population

t = time

k = growth rate

N = carry capacity

a = constant number of the fish harvested each time

The equation above represents the fish population with constant harvesting at a rate of $a=a_1$ wherein the effect of the initial condition decreases in the population of fish per season of harvesting.

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a$$

By separation of variables,

$$\int \frac{dP}{kP\left(1 - \frac{P}{N}\right) - a} = \int dt$$

Integrating left-hand side using partial fractions,

$$\int \frac{dP}{kP\left(1 - \frac{P}{N}\right) - a} = \int \left(\frac{A}{P - P_1} - \frac{B}{P - P_2}\right) dP$$

Substituting and integrating both sides,

$$\int \left(\frac{A}{P - P_1} - \frac{B}{P - P_2}\right) dP = \int dt$$

$$A\ln(P - P_1) + B\ln(P - P_2) = t + C$$

$$\ln\left((P - P_1)^A (P - P_2)^B\right) = t + C$$

Taking the anti-logarithm gives,

$$(P-P_1)^A(P-P_2)^B = Ce^t$$

Where P_1 and P_2 are roots to the quadratic polynomial P and $A=(P_1-P_2)^{-1}$, $B=(P_2-P_1)^{-1}$. At $P(0)= P_0$, then $C=(P_0-P_1)^A(P_0-P_2)^B$.

To get equilibrium solutions,

$$P(t) = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$P(t) = \frac{-(-N) \pm \sqrt{N^2 - 4(1)\left(\frac{aN}{k}\right)}}{2(1)}$$

$$P_1(t) = \frac{N + \sqrt{N^2 - 4\left(\frac{aN}{k}\right)}}{2}$$

$$P_2(t) = \frac{N - \sqrt{N^2 - 4\left(\frac{aN}{k}\right)}}{2}$$

This formula leads to the solution of roots of the equilibrium points.

B. Logistic Growth with Periodic Harvesting

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a(1 + sinbt)$$

The equation above is used in periodic harvesting wherein the parameter is set when a= a_2 there is a rapid drop of fish population.

In the sinusoidal equation $a(1 + \sin bt)$ parameter a represents the contribution of the sine wave to the growth function or the coefficient that determines the total rate of periodic harvesting. Parameter b represents the wavelength of the sinusoidal function that dictates periodic harvesting with respect to time (t). In this equation, b is equal to one season of harvesting.

III. Results and Discussions

A. Logistic Growth with Constant Harvesting

We assume that a constant number, a, of fish are removed from the population. Thus the logistic differential equation is given by,

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a$$

Where: P = size of the fish population

t = time

k = growth rate

N = carry capacity

a = constant number of the fish harvested each time

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a$$

By separation of variables,

$$\int \frac{dP}{kP\left(1 - \frac{P}{N}\right) - a} = \int dt$$

Integrating left-hand side using partial fractions,

$$\int \frac{dP}{kP\left(1 - \frac{P}{N}\right) - a} = \int \left(\frac{A}{P - P_1} - \frac{B}{P - P_2}\right) dP$$

Substituting and integrating both sides,

$$\int \left(\frac{A}{P - P_1} - \frac{B}{P - P_2}\right) dP = \int dt$$

$$A\ln(P - P_1) + B\ln(P - P_2) = t + C$$

$$\ln\left((P - P_1)^A (P - P_2)^B\right) = t + C$$

Taking the anti-logarithm gives,

$$(P - P_1)^A (P - P_2)^B = Ce^t$$

Where P_1 and P_2 are roots to the quadratic polynomial P and $A=(P_1-P_2)^{-1}$, $B=(P_2-P_1)^{-1}$. At $P(0)= P_0$, then $C=(P_0-P_1)^A(P_0-P_2)^B$.

In solving the differential equations, we use special techniques to solve equilibrium points

If $\frac{dP}{dt}=0$;

$$0=kP\left(1-\frac{P}{N}\right)-a$$

$$0=\frac{-N}{P}\left(kP+\frac{kP^2}{N}+a\right)$$

$$0=P^2-NP+\frac{aN}{k}$$

Where a=1

b= -N

$$c=\frac{aN}{k}$$

To get equilibrium solutions,

$$P(t)=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

$$P(t)=\frac{-(-N)\pm\sqrt{N^2-4(1)\left(\frac{aN}{k}\right)}}{2(1)}$$

$$P_1(t)=\frac{N+\sqrt{N^2-4\left(\frac{aN}{k}\right)}}{2}$$

$$P_2(t)=\frac{N-\sqrt{N^2-4\left(\frac{aN}{k}\right)}}{2}$$

This formula leads to the solution of roots of the equilibrium points.

Results of the experiment were gathered and tabulated. Choice 3 indicate values of k, N, and a1 being the values of the parameters in solving the logistic model.

Table 1. Chosen initial condition and solved P1 and P2

Choice	k	N	a1	p1	p2
3	0.2	5	0	5	0
			0.21	3.5	1.5
			0.22	3.366025404	1.633974596
			0.23	3.207106781	1.792893219
			0.24	3	2
			0.25	2.5	2.5

From this table we consider choice 3, where k=0.20, N=2, and we set the values of a1 from 0.21 to 0.25 to explain that the maximum value limits up to 0.25 only and extending to this value, the population will be undefined. This implies that when harvesting is done very often the population of fish decreases.

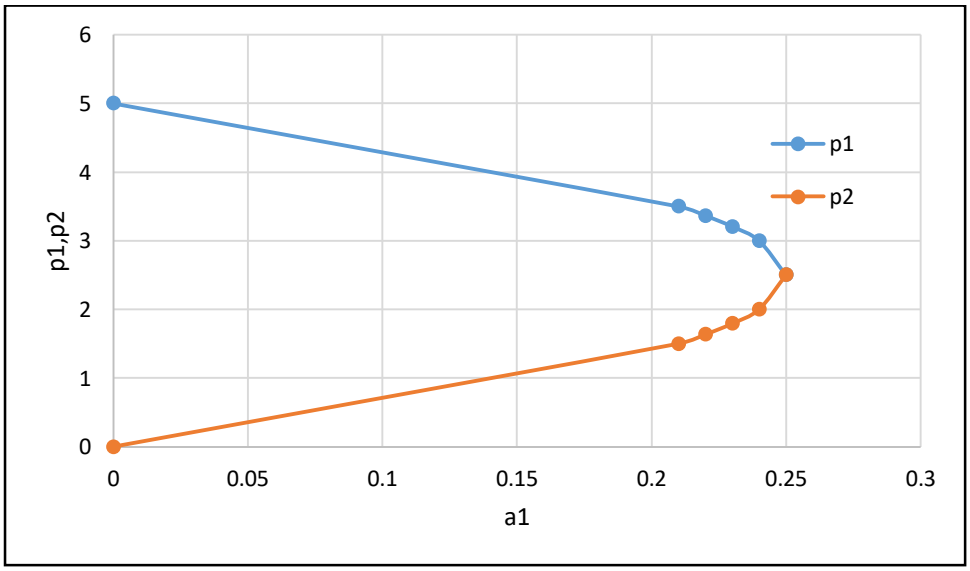

Figure 1. Graphical representation of table 1

Figure 1 shows the plotted points from table 1 which means that as we increase the value of a1 the equilibrium points meet at a certain point where a1 is equal to 0.25. At any initial population above the equilibrium point the population will decrease to the equilibrium value. At a1=0.25, the fish population and harvesting is stable and if the harvesting passes to the maximum value the fish population will die.

B. Logistic Growth with Periodic Harvesting

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a(1 + sinbt)$$

In the sinusoidal equation $\mathrm{a}(1 + \mathrm{sinbt})$ *parameter a represents the contribution of the sine wave to the growth function or the coefficient that determines the total rate of periodic harvesting. Parameter b represents the wavelength of the sinusoidal function that dictates periodic harvesting with respect to time (t). In this equation, b is equal to one season of harvesting.*

This equation is a non-autonomous and therefore difficult to separate. There is no clear way to solve this differential equation analytically because of the function of sine wave.

Though there is no clear analytic solutions to this differential equation we use qualitative and numerical techniques to the problem. So we use the aid of DE Tools technology to find solutions and graph for this study.

We note also that the solutions of non-autonomous equations are steady-state solutions that leads to the long term behaviour of the fish population.

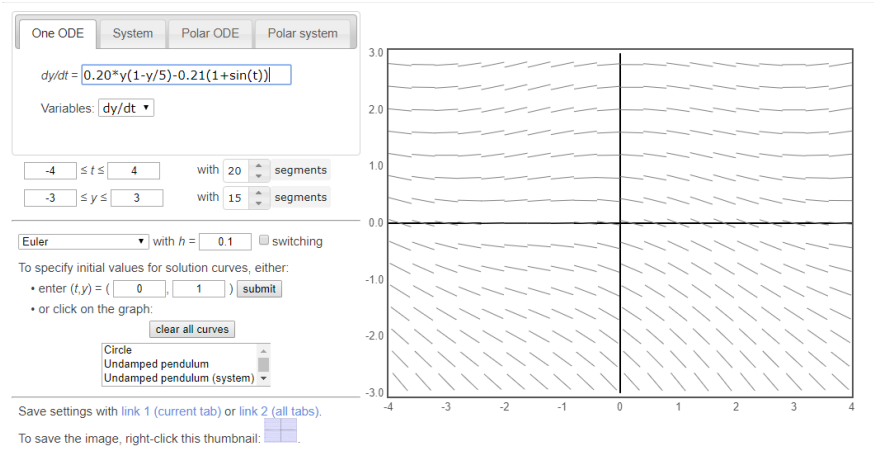

Figure 2. Slope field of the equation

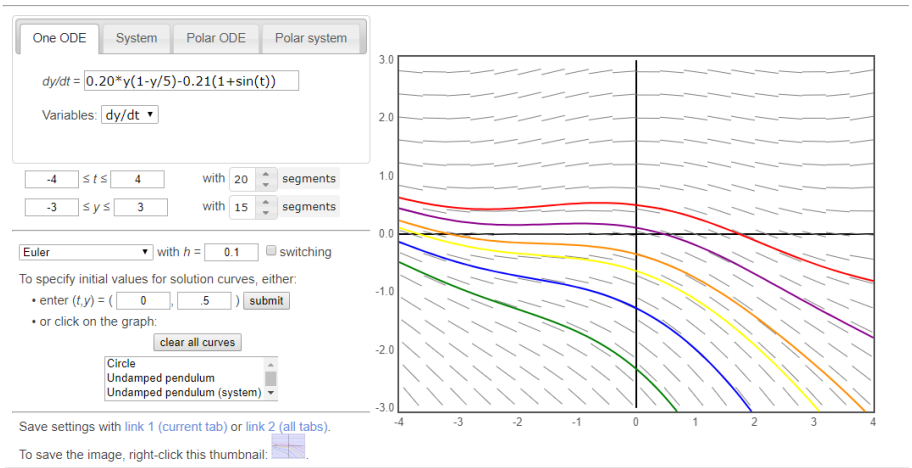

Figure 2.1 Phase line of the equation

Figure 2.1 shows the phase line of the given differential equation and its different values, it can be observed that this phase line are wave functions which means that the fish population is rapidly decreasing.

At a=a_2, with the same chosen parameters the differential equation is

$$\frac{dP}{dt} = 0.20P\left(1 - \frac{P}{N}\right) - 0.25(1 + sin\ (t))$$

Below are the results upon substituting the equation in the said technology.

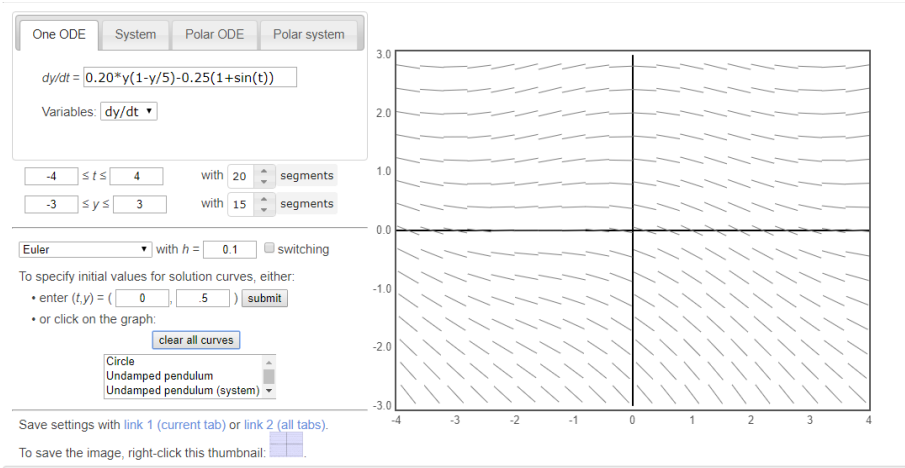

Figure 3. Slope field of the equation

Figure 3 shows the slope field of the equation to the given differential above with the value $a=a_2$.

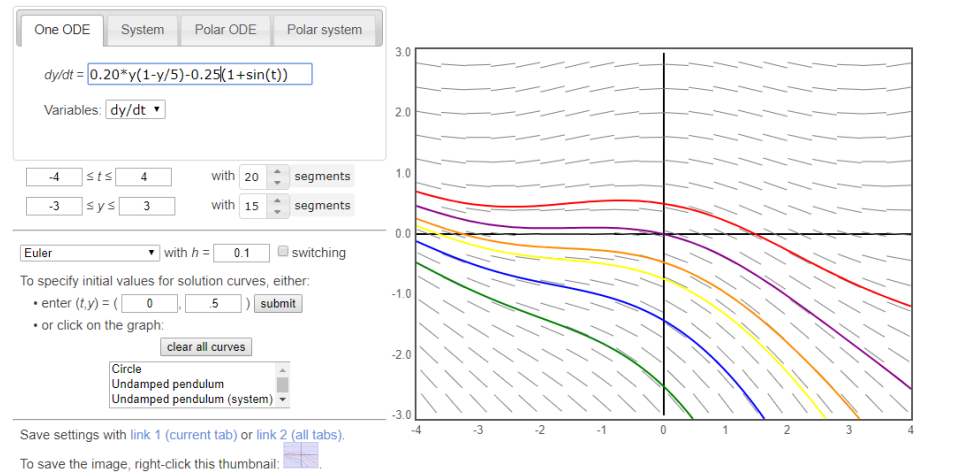

Figure 3.1 Phase line of the equation

Figure 3.1 show the phase lines of the solution and its different values. At this point the effect of the sine function are apparent in the slope field. Moreover, the sine function shifts the equilibrium state below the carrying capacity.

IV. Conclusion

This experiment hopes to seek the rate at which a person memorized a given data. To obtain this purpose, this experiment used a differential equation model of memorization to find the value of its k and eventually calculating the L (t).

In this experiment, two models were used to present the population harvesting, the constant harvesting and the periodic harvesting. In constant harvesting system, the values of a_1 are set from 0 to 0.25 to limit the system. Beyond the 0.25 value of a_1 the population of fish decreases until it approaches extinction. For periodic harvesting, since it is non-autonomous equation, the effect of the initial condition when $a=a_1$ is a decrease in the population of fish per season of harvesting. When $a= a_2$ there is a rapid drop of fish population.

Using the two models, logistic growth of periodic harvesting indicates better interval harvesting to maximize the fish population and to avoid its extinction.

V. References

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